

			Document number
Specification For Approval			
客户名称: Customer:	ShenZhen XinYuan Electronic Technology		
供应商名称: Customer:	Dongguan Honghui Electronics Co.		
Customer Code:	YLH100180		
Description:	2DB-915 Black	models	
Production: Zhou	Structure: Zhou		
	dates: 2024.07.25		

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I. Product Pictures



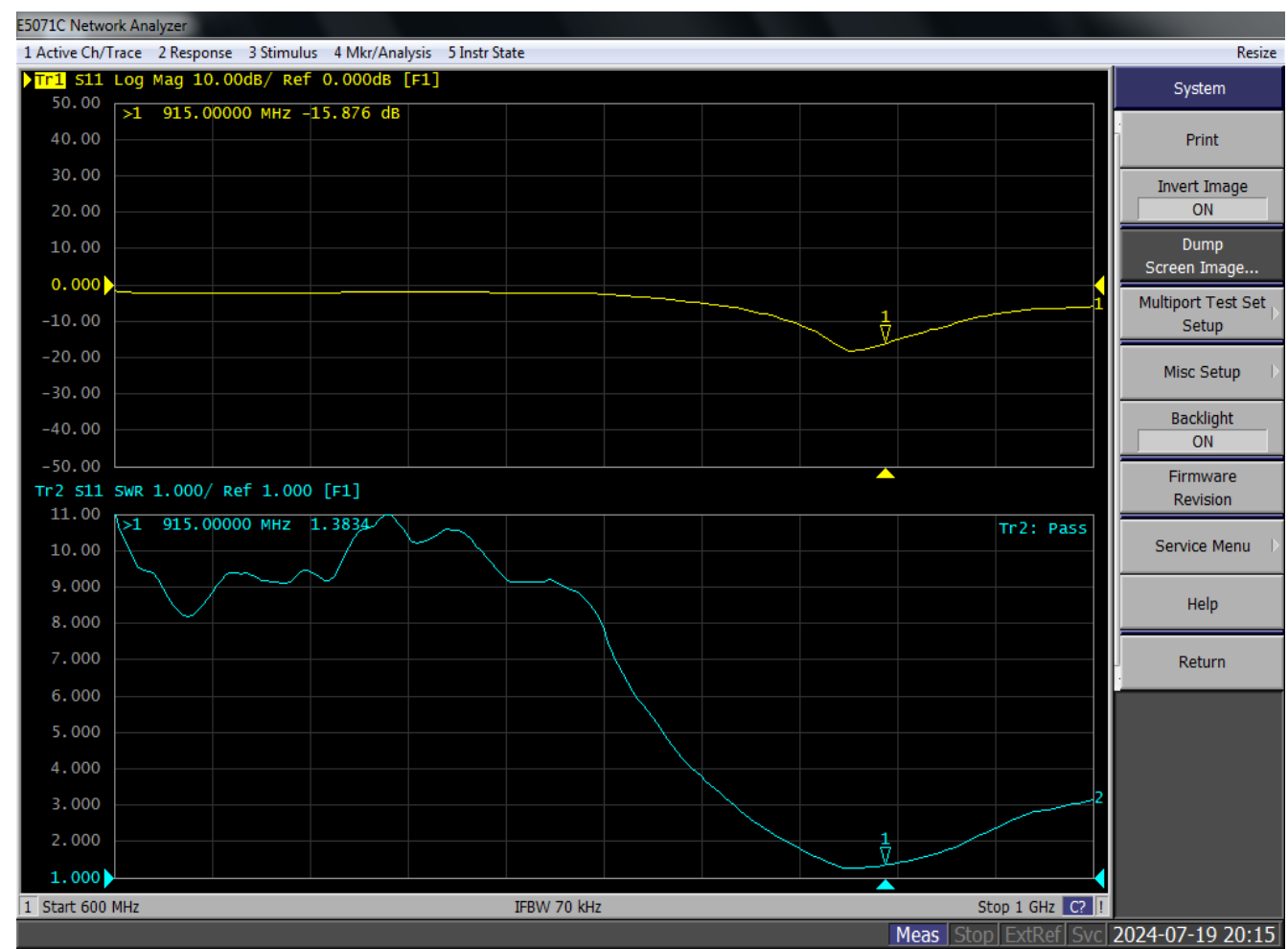
II. Product Parameters

Product Test Parameters			
产品名称(Name)	2DB-915 Black Antenna	产品型号(Model Type)	
电 性 能 指 标 (Electrical Specifications)			
频 率 范 围 (Frenquency Range)	915Mhz	极化方式 (Polarization)	perpendicular
输入阻抗(Impedance)	50 Ω	辐射方向	all directions
驻波比 (VSWR)	≤2	功率容量(Power)	50W
增益(Gain)	2dBi	带宽 (Bandwidth	100MHz

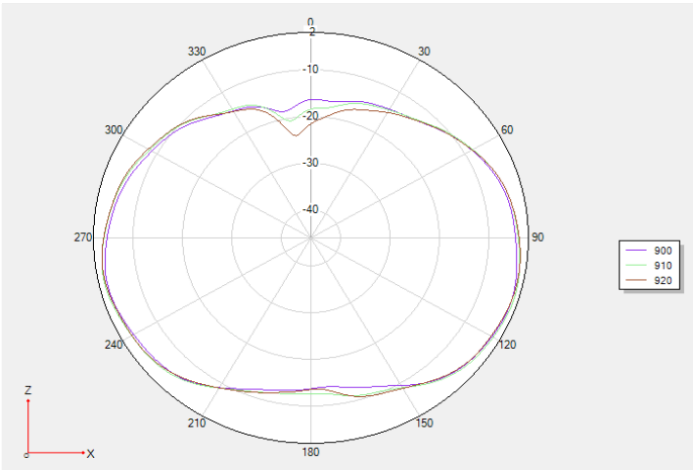
机械指标 (Mechanical Specifications)			
尺寸 (Dimensions)	88±2MM	天线颜色(Chassis Color)	black
连接器型号 (Connector)	SMA铜头母针	引线长度(Cable Length)	
天线材料 (Chassis Material)	ABS		
工作温度 (Working Temperature)	-40℃±85℃	存放温度 (Limit Temperature)	-40℃±85℃

III. S11 (VSWR 、 Return loss 、 Smith) Data 2D-3D Orientation Chart

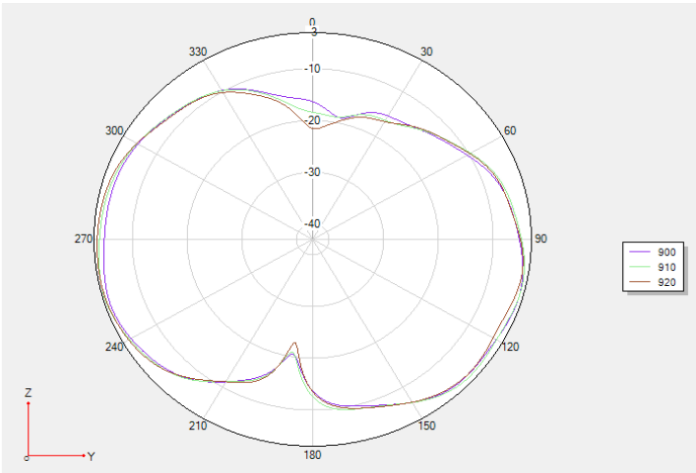
III-I. Network Analyzer Report



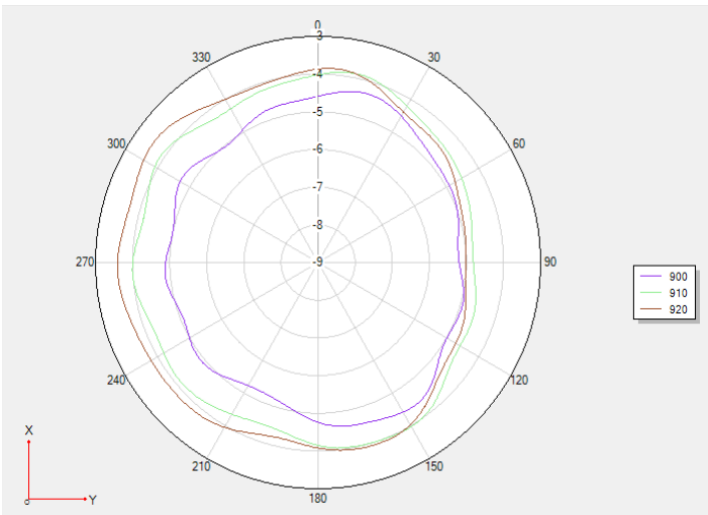
III-II. 2D-3D Direction Map



0°C

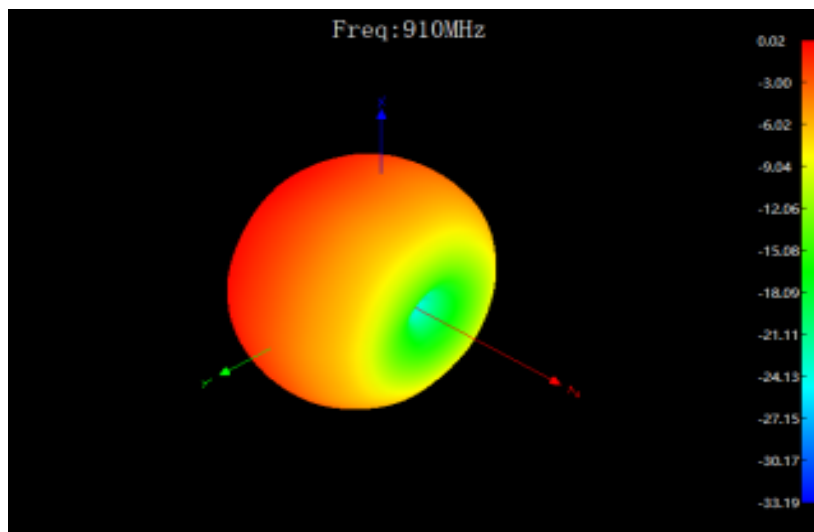


90°C

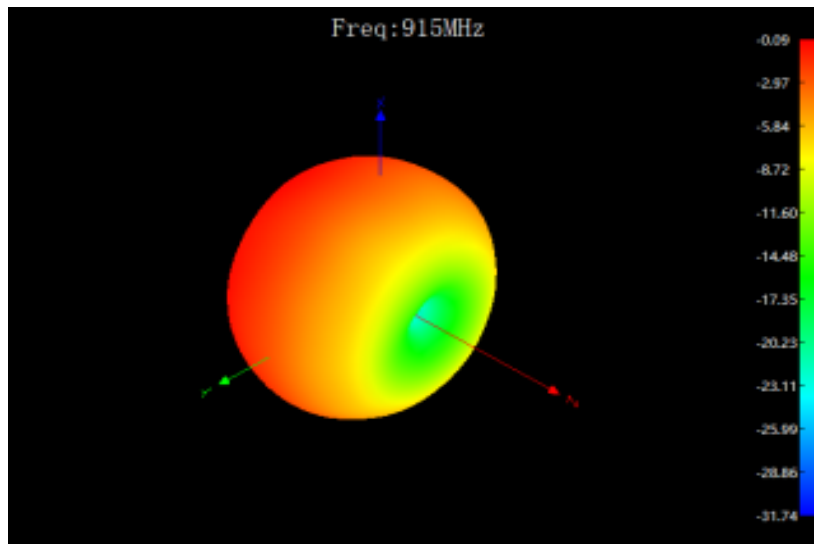


90°C

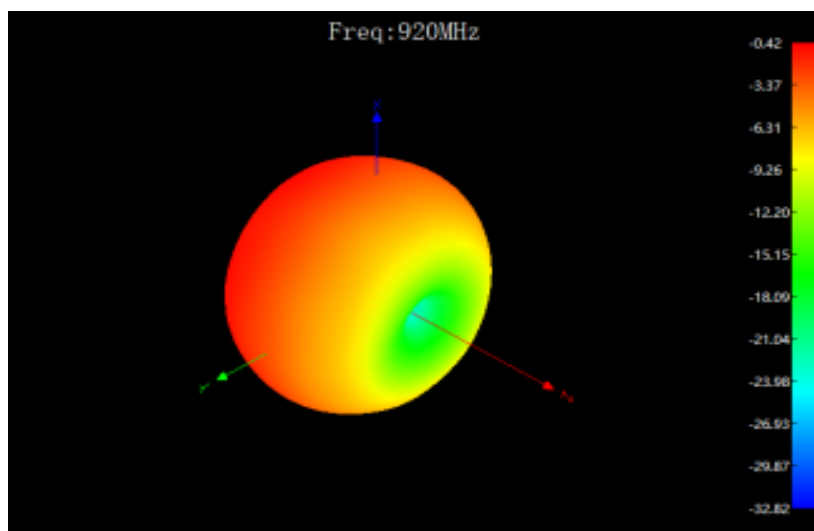
III-II. 2D-3D Direction Map



900MHz



915MHz



920MHz

IV. Antenna efficiency, gain value

[illegible]

Technical drawing of a SMA probe. The drawing includes a side view and a cross-sectional view. The side view shows a probe with a total length of 108.00 ± 2 mm. The probe has a diameter of 7.8 mm at the base and a diameter of 10.0 mm at the SMA head. The SMA head is labeled ② SMA铜头公针. The probe is labeled ① 胶棒外壳黑色. The cross-sectional view shows the internal structure of the probe, including a central pin and a base with a diameter of 29.26 mm.

LENGTH	A	B	C
L ≤ 10	±0.05	±0.10	±0.15
10 < L ≤ 50	±0.10	±0.15	±0.20
50 < L ≤ 150	±0.15	±0.20	±0.30
L > 150	±0.20	±0.25	±0.40

客户料号	料号	设计

版本	变更内容	日期	变更者	确认

角度公差	±1°	公差等级	B	单位	MM	第三视角

VI. Environmental performance testing

Storage environment	Test temperature, humidity, and barometric pressure without specifying as follows: 1. Temperature $-30^{\circ}\text{C}\sim+80^{\circ}\text{C}$. 2. Relative humidity of 45%-85 3. air pressure 86kpa-106kpa	Normal
High and low temperature test	5 cycles between 70°C and 40°C , followed by 1-2 hours under normal conditions to check the quality of the appearance.	Normal
Resistant to constant humidity and heat test	Relative humidity $95\pm 3\%$, test temperature: 40°C . Continuously 2H after the action, the test piece is taken out within 5min to determine the electrical properties, the test piece in the normal conditions of 1-2H, check the appearance of quality	Normal
Vibration test	Frequency range 10-55Hz, displacement amplitude: 0.35MM, acceleration amplitude: 50.0M/S, frequency sweep cycle times: 30 times	Normal
Drop test	3 free falls on mutually perpendicular axes at a height of 1M.	Normal